ORGANIZED BY DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING, CHANDIGARH UNIVERSITY

URBANSYNC AI

Autonomous Civic Operations, Mobility Mesh & 3D Spatial Digital Twin

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Team Introduction & Contributions

Chandigarh University engineering squad bridging high-throughput APIs, edge computer vision, and spatial WebGL.

AK **Akshat Kaushik**
FULL-STACK & APIS
async API architecture, Redis integration, frontend state

AT **Aaditya Tomar**
AI/ML & COMPUTER VISION
YOLOv8 fine-tuning, homography, edge inference

AY **Aditya Yadav**
GEOSPATIAL & 3D WEBGL
React Three Fiber digital twin, Deck.gl layers

AK **Aayush Kaushik**
CLOUD & DEPLOYMENT
Docker, sub-50ms WebSockets, CI/CD, PostGIS

Problem Statement

Urban Growth Fractures Civic Infrastructure into Blind Silos

THE CORE CRISIS

Rapid urbanization forces Sanitation, Mobility, and Emergency Services into isolated operational silos, causing delayed dispatches, unmonitored waste accumulation, and fuel waste.

TARGET AUDIENCE

Municipal Corporations, City Traffic Police, Emergency First Responders (Ambulance/Fire), and Daily Urban Commuters.

WHY IT MATTERS

Fragmented civic data causes life-threatening emergency corridor delays, severe neighborhood waste accumulation, and uncoordinated traffic gridlocks.



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Problem Context & Existing Solutions

Current smart city deployments fail due to hardware fragility, theft, and isolated data architectures.

EXISTING SMART CITY IOT

Physical Hardware Traps

Rely on physical ultrasonic bin sensors and road magnetometer pucks prone to vandalism, theft, weathering, and recurrent battery maintenance costs.

CURRENT OPERATIONAL STATE

Fragmented Applications

Isolated point tools: standalone parking apps, fixed-schedule garbage collection routes, and manual officer traffic signal overriding.

IDENTIFIED GAPS

Zero Real-Time Mesh

No cross-departmental synchronization, total dependency on expensive field hardware, and complete absence of dynamic real-time spatial optimization.



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UrbanSync AI: A Hardware-Free Municipal Command Platform

Zero Physical Sensors. Purely Software-Driven Intelligence.

Overview

Purely software-driven, hardware-free municipal command platform.

Core idea

Synchronizes departments through event-driven pub/sub, dynamic algorithmic routing, edge-anonymized computer vision and an interactive 3D WebGL digital twin.

How it helps

Turns existing static camera feeds into actionable data, routes sanitation trucks by fill-volume, and preemptively clears traffic corridors for emergency vehicles.



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Key Features & Innovation

Proprietary algorithmic pipelines delivering civilian safety, dynamic logistics, and automated traffic dispatching.

01
SPATIAL TWIN
3D Digital Twin
Real-time WebGL command interface visualizing traffic density, waste hotspots, and fleet status.

02
SAFETY MESH
SafePath Routing
Nirbhaya-Mesh for women's safety using 3D occlusion raycasting and dynamic street lighting analysis.

03
TRAFFIC ENGINE
SILN Corridor
Dynamic departure scheduling with preemptive green-wave corridor clearance for emergency units.

04
PRIVACY SHIELD
Privacy-by-Design
Gaussian facial and license plate blurring at edge ingestion with strict zero-biometric storage.

05
AI INNOVATION
Homography & OR-Tools
Perspective alignment for waste delta tracking (T0 vs T1) paired with Google OR-Tools multi-truck solver.

Proposed Workflow

High-velocity streaming data pipeline connecting edge vision, in-memory brokerage, and distributed spatial actors.

Synthetic Ingestion Layer

Ingestion: Analyzes CCTV feeds and citizen telemetry

In-Memory Message Broker

Processing: Evaluates waste delta and traffic nodes

Core Optimization Engines

Action: Updates Redis Pub/Sub channels

Persistent Spatial Storage

Spatial storage (PostGIS spatial tables)

DISPATCHERS
See real-time 3D alerts.

FIELD WORKERS · PWA
Get optimized routes via PWA.

CITIZENS
Get safe navigation paths.

Proposed Tech Stack

Battle-tested open-source frameworks, high-throughput asynchronous backends, and accelerated WebGL graphics.

FRONTEND

Next.js 14

React Three Fiber, Three.js, Deck.gl spatial layers, and Tailwind CSS.

BACKEND

FastAPI

Python 3.11+, Async REST endpoints, and high-frequency WebSocket streams.

DATABASE

PostgreSQL

PostGIS spatial extensions paired with Redis 7.x in-memory Pub/Sub cache.

AI / CV

YOLOv8 & OpenCV

ONNX/PyTorch inference, Homography matrices, and Google OR-Tools CVRP solver.

SPATIAL APIS

NetworkX & Mapbox

OSMnx for directed graph routing, OpenStreetMap data, and Mapbox GL shaders.

Integration Architecture: Asynchronous REST APIs handle administrative triggers, while dedicated WebSockets push live vehicle telemetry and raycast safety maps to client devices.

Feasibility & Execution Plan

Approach: Microservices architecture deployed via Docker

Milestones

1. Core API & database modeling
2. CV model integration and OR-Tools setup
3. 3D spatial dashboard integration
4. Real-time testing via simulation daemons

Challenges

- ▲ High-frequency WebSocket streams
- ▲ Sub-50ms event sync
- ▲ Accurate 3D spatial rendering

Strategy: Parallel development of ML engines and Next.js frontend, unified through Docker-compose orchestration.

Expected Impact & Future Scope

Transforming municipal operations with proven quantifiable metrics and infinite modular scalability.

IMPACT

30% faster

emergency response

IMPACT

Lower

municipal carbon footprint

IMPACT

Safer

urban transit

BENEFITS

Major cost savings

For city councils by eliminating IoT hardware maintenance.

SCALABILITY

Any municipal ward

Mapping OpenStreetMap data and connecting IP camera feeds.

FUTURE

Expanded civic ops

Drone-based feed integration, automated ticketing via E-Challan.



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